

Project Proposal: Predicting Medical Insurance Charges

Introduction

Medical insurance costs can be very different from one person to another, and both insurance companies and customers can benefit from understanding which personal factors make these costs go up, so that prices are fair and people can make healthier choices. In this project I plan to build a regression model that predicts a person's yearly insurance charges and shows which factors, such as age, BMI and smoking, have the biggest effect on how much they pay.

Dataset

I will use the Medical Cost Personal Dataset (Lantz, 2013), a CSV file with 1,338 rows and 7 columns that describes simulated US insurance customers, where each row is one person and the target variable, charges, is a continuous number in US dollars that ranges from about \$1,122 to \$63,770. The six predictors include three numeric variables (age, BMI and number of children) and three categorical variables (sex, smoker status and region), which gives a good mix of variable types and should make it possible to see how each one affects cost.

Data cleaning

To prepare the data, I will first check for missing values and remove any duplicate rows so that no person is counted twice, then convert the categorical variables into numeric 0/1 dummy variables so they can be used in a regression model, and finally check numeric columns such as BMI for unrealistic outliers that could be data entry errors.

Initial visualizations

To understand the data, I will start with a histogram of charges to see whether the target is skewed, and then use scatter plots of charges against age and BMI, coloured by smoker status, along with box plots of charges for each categorical variable, to see how each predictor relates to cost and whether some groups pay much more than others. A correlation heatmap will then summarize how strongly each variable is related to charges and flag any predictors that are strongly correlated with each other, which could cause problems for a linear model.

Regression model

I will use multiple linear regression in Python (pandas, scikit-learn, statsmodels), because the target is continuous, there are several predictors, and the coefficients are easy to interpret, so I can see how much each factor changes the cost while keeping the others the same. I will split the data into a training set (80%) and a test set (20%) and evaluate the model with R^2 , RMSE and MAE, as well as residual plots, to check how well it predicts charges for people it has not seen before. Because charges are likely right-skewed and some relationships may not be straight lines, I will also test a log-transformed target and consider adding terms such as age squared or interactions between variables, and then compare these models to choose the one that performs best.

References

Lantz, B. (2013). *Machine learning with R*. Packt Publishing.

Stedy. (n.d.). *Machine-Learning-with-R-datasets* [Data set]. GitHub. Retrieved October 3, 2026, from <https://github.com/stedy/Machine-Learning-with-R-datasets>